Collinearity and Cluster Stability in Transport Economics: Evidence from Aviation and Rail

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Abstract

Collinearity is endemic in transport performance data, yet its effect on k-means cluster assignments—whether to cluster in raw variable space or in PCA score space—has not been systematically evaluated. We show that the answer depends critically on the *source* of variation. In a time-series setting (26 annual US airline observations, 1995–2020), collinearity compresses variance onto a secular growth axis but preserves all cluster assignments perfectly (ARI $= 1.000$ for $k = 2$–9), and the data structurally supports three competitive eras. In a cross-sectional setting (17 European rail operators, UIC RAILISA, 1995–2024), between-operator structural heterogeneity raises effective dimensionality (participation ratio $= 4.00$ vs. 2.47 for aviation) and makes PCA dimensionality choices consequential: full assignment recovery requires retaining four PCs, and breaks down at $k \geq 6$. Across both studies, the optimal cluster count approximates the participation ratio ($k^* = 3 \approx \lfloor 2.47 \rfloor$; $k^* = 4 = \lfloor 4.00 \rfloor$), suggesting the participation ratio as a principled prior for cluster-count selection. A six-kernel PCA battery confirms a linear cluster manifold in aviation and yields inconclusive but suggestive nonlinear structure in rail.

Keywords: transport economics; k-means clustering; principal component analysis; kernel PCA; multicollinearity; effective dimensionality; airline profitability; European rail; participation ratio

1.0 Introduction

Transport performance datasets share a well-known structural property: passenger volumes, load factors, revenues, and cost ratios co-move with network growth, creating severe collinearity

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among the variables researchers use to cluster operators or time periods. Two responses appear in the literature: ignore the collinearity and cluster in raw variable space [Renold et al., 2023], or decorrelate via Principal Component Analysis (PCA) before clustering. Neither approach has been systematically evaluated—and the right answer turns out to depend on a dimension that has received no attention: whether the source of variation is *within-unit* (shared time trend) or *between-unit* (structural differences across operators or countries).

This paper examines both cases using two transport datasets with parallel variable structures. **Study 1** replicates and extends Renold et al. [2023], who cluster 26 annual US airline observations (1995–2020) on seven variables. Their key finding—that COVID-19 does not permanently disrupt airline profit regimes—is sound, but two methodological questions are left unexamined: does collinearity distort cluster assignments, and what k does the data structurally support? **Study 2** extends the framework to 17 European rail operators (UIC RAILISA, 1995–2024) using seven directly analogous variables. Here the clustering problem is structurally different: operators differ in liberalization regime, subsidy dependence, and market structure—dimensions that cut *across* time rather than along it. This cross-sectional variation makes collinearity consequential in a way the time-series case does not.

Three research questions unify the two studies:

(i) Does collinearity distort k-means cluster assignments when moving from raw variable space to PC score space, and does the answer differ between time-series and cross-sectional settings?

(ii) What number of clusters does each dataset structurally support, and is this signal suppressed by collinearity?

(iii) Is the cluster manifold intrinsically linear, or does nonlinear structure emerge—particularly in the cross-sectional case where regime boundaries may be sharp?

2.0 Theoretical Framework

2.1 Collinearity geometry and the participation ratio

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a standardized data matrix with eigenvalue spectrum $\lambda_1 \geq \cdots \geq \lambda_p > 0$. The *participation ratio*

$$PR = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \tag{1}$$

measures effective dimensionality on a scale from 1 (single dominant factor) to p (all dimensions equal). The formula is the reciprocal of the Herfindahl concentration index applied to the eigenvalue spectrum; it is equivalent to the "effective rank" of Roy and Vetterli [2007] and has been used as a dimensionality diagnostic across the signal processing and machine learning literature. Severe collinearity yields $PR \ll p$.

2.2 When collinearity preserves cluster assignments

In a time-series setting where a single secular trend dominates ($\lambda_1 \gg \lambda_2$), all p variables load heavily on PC1. k-means in raw space inflates distances along PC1 by $\sim \sqrt{p \cdot \lambda_1 / \lambda_j}$ relative to each minor axis, but does not reorder points along PC1. If cluster structure is *chronological*—each cluster is a contiguous time band along the growth axis—then stretching the temporal axis preserves band boundaries. Assignment robustness (ARI ≈ 1) follows directly.

2.3 When collinearity distorts cluster assignments

In a cross-sectional setting, units differ structurally: some dimensions reflect the within-unit trend (shared by all operators) while others reflect between-unit regime differences. If a regime dimension has low variance marginally but high discriminative power for cluster membership, collinearity suppresses its relative weight in raw space. k-means then over-weights the shared growth axis and under-weights the regime axis, potentially merging clusters that PCA would separate. Assignment distortion (ARI < 1) is the expected outcome.

2.4 Kernel PCA as nonlinearity diagnostic

Standard PCA assumes cluster boundaries are linear in the input space. If regime boundaries are sharp (e.g., liberalized versus state monopoly), a nonlinear kernel may resolve ambiguous assignments that linear PCA cannot. We apply six kernels spanning three families: distance-based (RBF, poly-2, poly-3), information-geometric (Fisher–Mahalanobis), and graph-based (diffusion heat kernel on the 5-NN graph). Consensus across all six kernels constitutes strong evidence for the linear case; divergence suggests nonlinear structure.

Figures 1 and 2 illustrate both cases schematically.

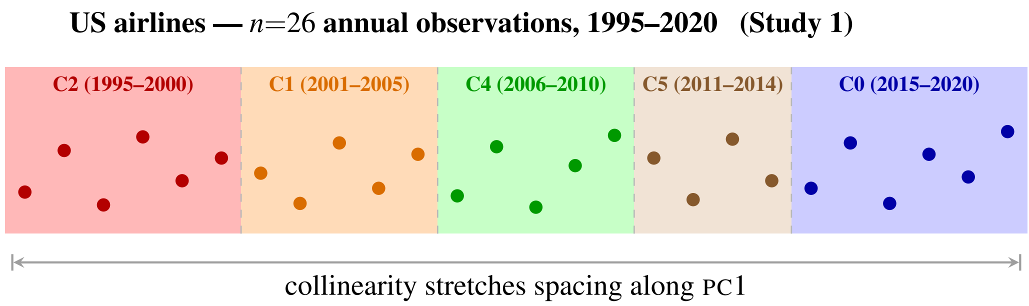

Figure 1: Within-unit (time-series): Clusters are contiguous temporal bands along PC1. Band widths are proportional to cluster size (6/5/5/4/6 years). Collinearity stretches distances along PC1 but cannot reorder points: boundaries perpendicular to PC1 survive any uniform scaling (ARI = 1.000, all k, Study 1).

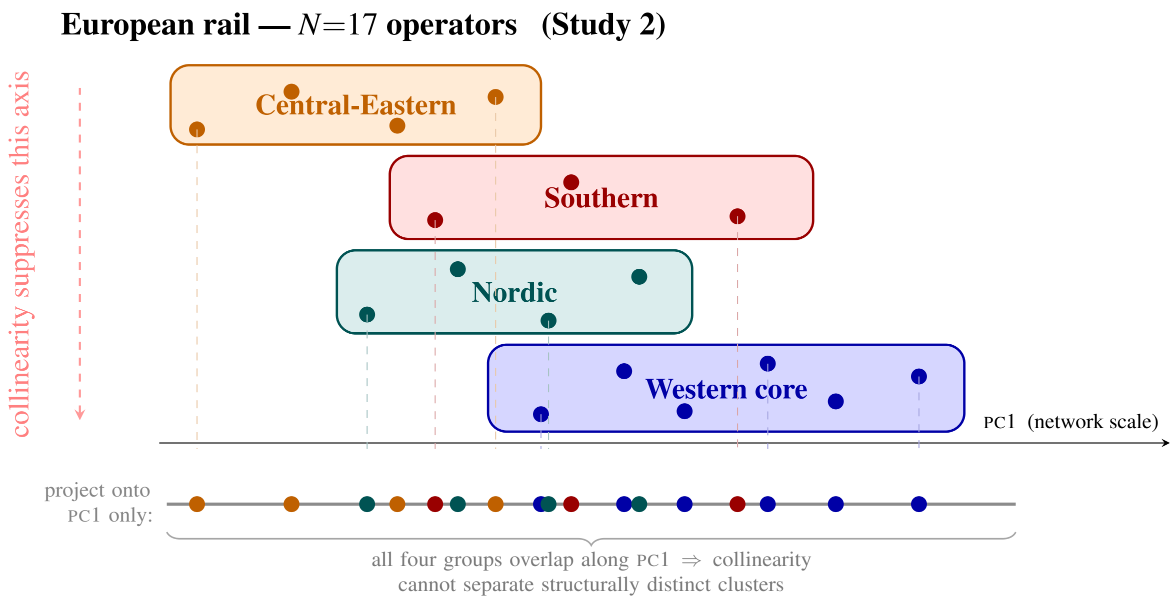

Figure 2: Between-unit (cross-section): Four regime groups (Western core, Nordic, Southern, Central-Eastern) are well-separated along the regime axis (red, vertical) but heavily overlapping along PC1 (horizontal). Raw-space collinearity suppresses the regime axis (dashed arrow, left), leaving only PC1 visible to k-means. The grey strip shows all operators projected onto PC1 alone: groups interleave completely, distorting assignments (ARI $<$ 1.000 at $k > k^*$, Study 2).

3.0 Shared Methodology

Clustering. k-means with 100 random initializations and up to 1000 iterations per run. Three spaces compared in each study: (a) standardized raw variables (pD-raw), (b) PC1–PC3 scores (3D-PC), (c) PC1–PC4 scores (4D-PC). Assignment agreement measured by ARI [Hubert and Arabie, 1985].

Cluster quality. Silhouette coefficient *Sil* [Rousseeuw, 1987] and Davies–Bouldin index *DB* [Davies and Bouldin, 1979], evaluated for $k = 2, \ldots, 9$.

Kernel PCA battery. Applied to 4D-PC score vectors. Bandwidths set via median pairwise distance heuristic (principled, no cross-validation). Assignment agreement between KPCA-projected partitions and linear PCA partitions measured by ARI.

Implementation. Python 3.12, scikit-learn 1.x. All results reproducible from publicly available data (Study 1: Renold et al. 2023 Table 4; Study 2: UIC RAILISA).

4.0 Study 1: US Airline Profitability Cycles (Time Series)

4.1 Data

Analysis uses 26 annual observations (1995–2020) from Renold et al. [2023]'s Table 4: operating profit, fuel price, average wages, revenue passenger miles (RPM), load factor, yield per passenger, and other expenses ratio. All variables standardized (zero mean, unit variance) before distance computations.

4.2 PCA structure

Table 1 reports the eigenvalue spectrum. PC1 captures 59.8% of variance with uniformly positive loadings across all seven variables (Table 2): the fingerprint of a single secular growth factor. The participation ratio $PR \approx 2.47$ confirms severe collinearity: seven variables carry effectively 2.47 independent dimensions of information.

PC axes as strategic lenses. PC1 functions as the industry-cycle clock: since it captures secular growth common to all seven variables, the year of a benchmarking exercise matters as much as the identity of the peer group. A carrier benchmarked in 1998 and one benchmarked in 2018 are operating in structurally different environments, even if both appear in the same raw-space cluster. PC2 represents the core strategic tension in commercial aviation: it contrasts carriers that grow revenue (positive loading on operating profit) against those that control costs through wage discipline (negative loading on wages). Strategy choices along this axis are the primary differentiator between carriers within a given era. PC3 is the fuel-price exposure axis, capturing the divergence between yield-per-passenger and fuel costs. Carriers with aggressive fuel hedging programmes occupy a distinct region of PC3 space relative to unhedged carriers, explaining why identical revenue growth trajectories can produce different cluster assignments during oil-price shock years.

Table 1: Study 1 (Aviation): Variance explained by each principal component.

PC	Interpretation	Variance (%)	Cumulative (%)
1	Secular US aviation growth	59.8	59.8
2	Profit/wage boom–bust tension	16.7	76.5
3	Fuel-price shocks	11.3	87.8
4	COVID-era yield collapse	7.8	95.5
5–7	Noise	4.5	100.0

Table 2: Study 1 (Aviation): Loading scores on PC1–PC4. Bold: $|\text{loading}| > 0.40$.

	Variable						
PC	Op. Profit	Fuel	Wages	RPM	LF	Yield	Other Exp.
1 (59.8%)	+0.33	+0.39	+0.28	**+0.45**	**+0.45**	+0.34	−0.38
2 (16.7%)	**+0.63**	−0.30	**−0.51**	+0.23	+0.27	−0.22	+0.27
3 (11.3%)	+0.24	**−0.45**	+0.55	+0.08	−0.17	**+0.45**	+0.46
4 (7.8%)	+0.16	+0.17	−0.50	−0.37	−0.18	**+0.73**	−0.04

4.3 Finding S1-I: Cluster assignments are robust to collinearity

k-means in 3D-PC space produces identical assignments to 7D-raw space for all 26 observations (ARI = 1.000, 26/26 agreement). The mechanism: each of the six clusters is a *contiguous chronological band* along PC1. Collinearity inflates distances along PC1 but does not reorder points along it, so band boundaries are preserved. The 4D-PC space differs in a single assignment: the year 2000 observation (coinciding with the dot-com bust and an oil price spike) is reassigned away from the 9/11-era cluster, reflecting the PC4 yield-per-PAX axis distinguishing high-demand/high-fuel years from the post-9/11 demand collapse.

Business takeaway. The $\text{ARI} = 1.000$ result has a direct practical implication: airline consultancies can safely construct peer groups from raw KPI rankings without correcting for collinearity. The peer lists do not change when you apply PCA pre-processing. However, the year of the benchmark matters enormously. A carrier peer-grouped in 1998 belongs to the low-profit, post-deregulation era; the same carrier peer-grouped in 2018 belongs to the peak-profit era. Era effects dominate firm-level effects along PC1, which means that cross-era benchmarking—comparing a carrier's 2001 performance to 2019 industry norms, for example—is geometrically equivalent to comparing points from different clusters.

4.4 Finding S1-II: The data supports three clusters, not six

In geometrically clean 3D-PC space, the silhouette coefficient peaks decisively at $k = 3$ (0.523), while $k = 6$ scores 0.448—a 14% gap. In 7D-raw space the curve is flat between $k = 6$ and $k = 7$ (0.448 vs. 0.449) with no clear maximum: collinearity's inflated growth axis suppresses the between-cluster contrast that would identify $k = 3$. The three-cluster solution corresponds to low-profit era (1995–2005), recovery era (2006–2014), and peak-profit era (2015–2019), with 2020 as a structural singularity. The six-cluster taxonomy is a descriptively coherent subdivision of these three regimes, but is imposed on the data rather than recommended by it. Figure 3 summarizes the quality metrics visually.

Consulting frameworks and competitive eras. The $k^* = 3$ finding implies that US airlines have three competitive eras, not six strategic archetypes. Management consulting frameworks that divide airlines into more than three strategic groups—distinguishing, say, "network incumbents", "hybrid carriers", "ultra-low-cost carriers", "regional specialists", "Gulf challengers", and "legacy restructurers"— are subdividing within eras rather than across them. The six-cluster taxonomy of Renold et al. [2023] is descriptively coherent but reflects the time-period structure of the data, not a genuine six-way strategic differentiation. The 2020 singularity is a special case:

the COVID year is statistically isolated from all pre-COVID years in every clustering space tested. Comparing 2020 financial performance to any pre-COVID peer group is geometrically indefensible; the COVID observation is not a member of any pre-existing strategic cluster, but a structural shock that temporarily created its own singleton regime.

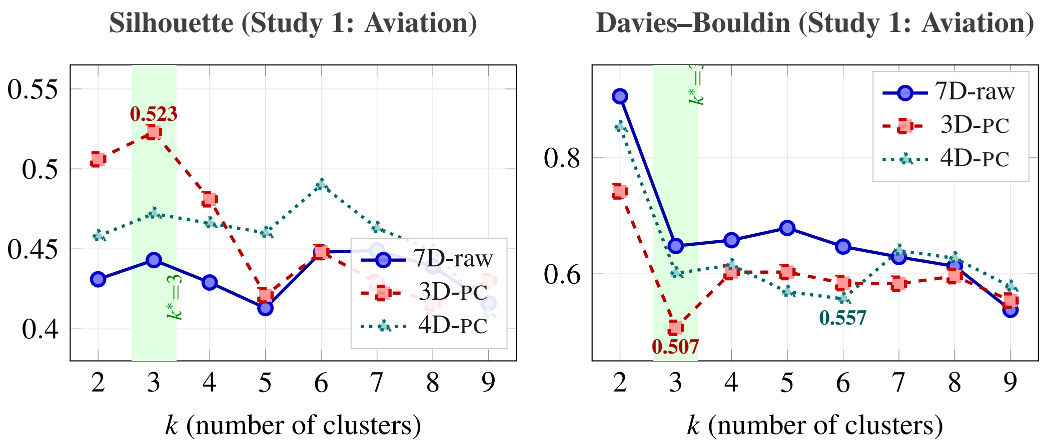

Figure 3: Study 1 (Aviation): Silhouette coefficients (left) and Davies–Bouldin indices (right) for $k = 2, \ldots, 9$ across three clustering spaces. Green band marks the structural optimum $k^* = 3$. In geometrically clean 3D-PC space (red dashed), both metrics identify $k = 3$: silhouette peaks at 0.523, Davies–Bouldin troughs at 0.507. In 7D-raw space (blue solid), collinearity flattens the silhouette curve and suppresses the $k^* = 3$ signal. The 4D-PC space (teal dotted) achieves the best DB index at $k = 6$ (0.557).

4.5 Finding S1-III: 4D-PC is the quantitatively superior space

The 4D-PC space achieves the best Davies–Bouldin index at $k = 6$ (0.557 vs. 0.647 in 7D-raw, a 13.9% improvement), adding PC4 which captures the COVID yield collapse.

Why PC4 unlocks the sixth cluster. The fact that PC4 is necessary for the best *DB* index at $k = 6$ reveals a specific geometric mechanism: the sixth cluster becomes justifiable only when the COVID year's extreme yield contraction is explicitly represented in the feature space. Without PC4, the 2020 observation bleeds into the peak-profit cluster along the growth axis, because its high load factor and revenue-passenger-mile trajectory superficially resemble those of the 2015–2019 era. PC4 provides the orthogonal axis that separates COVID's yield collapse from the structural characteristics of peak-profit years, making the sixth cluster geometrically real rather than an artefact of projecting a shock event onto pre-existing axes.

4.6 Finding S1-IV: Kernel PCA confirms linear manifold

All six kernel functions [Schölkopf et al., 1998] preserve the six-cluster assignment in 2D (ARI $= 1.000$ vs. linear PCA). In 1D projection, the linear kernel conflates COVID year C3 with peak-profit cluster C0 along the temporal axis; all five non-baseline kernels shift C3 to overlap post-financial-crisis cluster C5. Agreement across three fundamentally different kernel families (distance-based, information-geometric, graph-based) confirms an intrinsically linear manifold with no hidden curvature. The participation ratio $PR \approx 2.47$ predicted this; KPCA confirms it. Figure 4 shows the 1D projection for each kernel.

Implications for practitioners. Linear manifold confirmation carries a direct methodological guarantee: there is no hidden nonlinear structure in airline cost dynamics. The competitive landscape evolves smoothly along the growth trajectory encoded in PC1, with no sharp discontinuities between strategic regimes. The exception is COVID, which is a structural shock rather than a regime transition—it displaces the 2020 observation off the smooth manifold rather than warping the manifold itself. For M&A due diligence and strategic planning, this means historical airline financial data can be analysed safely with standard linear tools (PCA, discriminant analysis, linear regression): nonlinear methods add noise rather than signal in this domain.

5.0 Study 2: European Rail Operators (Cross-Section)

5.1 Data

Source. UIC RAILISA database [Union Internationale des Chemins de fer (UIC), 2024], covering ≥ 100 railway companies worldwide with continuous reporting from 1995. We extract seven variables in direct correspondence with Renold et al.'s aviation variables (Table 3).

All RAILISA variables used in this study—EBITDA, staff expenditure, material and external service costs, total operating expenditure, total operating revenue, passenger revenue, passenger-kilometres, and mean annual staff strength—are financial or operational figures that each operator discloses in its published annual report and statutory accounts. RAILISA provides a structured aggregation of these publicly reported figures; it is a convenience database, not

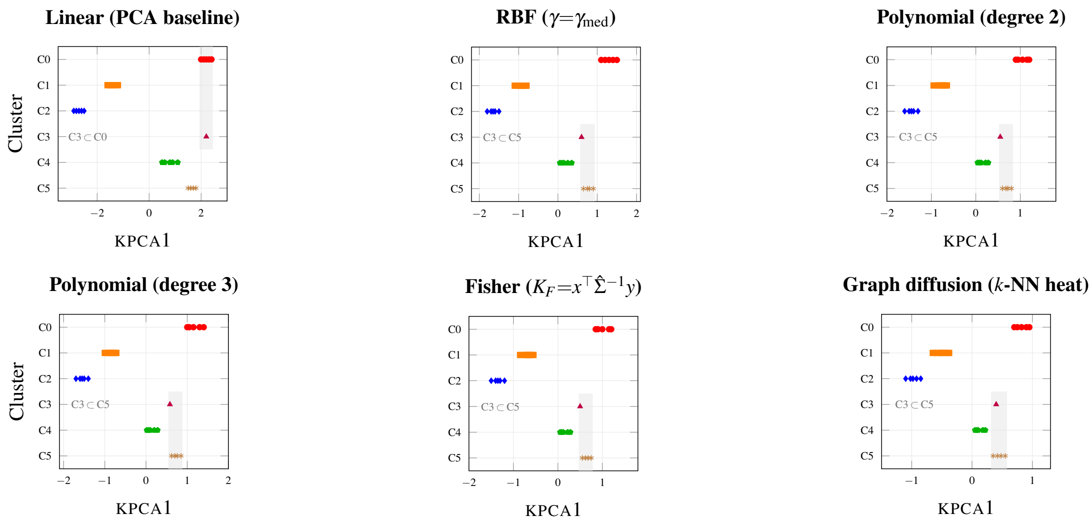

Figure 4: Study 1 (Aviation): 1D KPCA strip charts for six kernels spanning three families. Each row is a cluster; dots are the 26 observations projected onto KPCA1. Shaded bands mark regions where the two clusters cannot be separated in 1D. **Linear** (top-left): COVID year C3 (purple, +2.2) falls inside the C0 band—inseparable because PC1 dominates. **All five non-baseline kernels** (remaining panels): C3 shifts to ≈ 0.40–0.60, overlapping C5 instead of C0. The shift mechanism differs—curvature-fitting (RBF, poly), eigenvalue weighting (Fisher), graph connectivity (diffusion)—but all five agree on the same ambiguous pair. Consensus across three fundamentally different kernel families confirms an intrinsically linear manifold; KPCA2 resolves the C3–C5 ambiguity in every kernel, mirroring the role of PC2 in standard PCA.

a confidential data source. The RAILISA `value_visibility` flag distinguishes records that RAILISA's redistribution agreement designates as restricted from those it designates as public, but this flag reflects the terms of UIC's data-sharing arrangement with reporting companies, not the confidentiality of the underlying data. Every observation used in this analysis can be independently verified against the corresponding operator's published annual report for the relevant year. EUR conversion rates are ECB annual averages [European Central Bank, 2025], which are publicly available without restriction.

Operator sample. 17 European passenger operators selected for continuous RAILISA reporting since $\approx$1995, sufficient financial scale, and one entity per country (Table 4). The sample spans four regime types: Western core (commercially oriented), Nordic (state-owned, varying liberalization), Southern (high-speed investment, subsidy-dependent), and Central-Eastern (transitional, heavily subsidized).

Panel structure. Each of the 17 operators is represented by time-averaged values (1995–2024), giving $N = 17$ observations in 7D space. This isolates *between-operator* (structural) variation—

Table 3: Variable mapping between Study 1 (aviation) and Study 2 (rail). Rail variables constructed from RAILISA indicators as detailed.

Rail variable	Renold et al. analogue	Derivation	Unit
EBITDA margin	Operating profit margin	7225 / 7215	fraction
Cost recovery	Revenue / cost ratio	7223 / 7213	fraction
Staff ratio	Labour intensity	7209 / 7213	fraction
Material ratio	Material intensity	7206 / 7213	fraction
Staff productivity	RPM analogue	5113 / 3111	1000 pkm/FTE
Yield per pkm	Yield per passenger	EUR(7215) / 5113	EUR/1000 pkm
Network scale	RPM scale	log(5113)	log bn pkm

VAR_IDs: 7225 EBITDA; 7215 passenger revenue; 7223 total operating revenue; 7209 staff expenditure; 7206 material/external services; 7213 total OPEX; 5113 passenger-km; 3111 mean annual staff. Energy (8103–8108) and occupancy (5205, 5208) variables excluded due to incomplete coverage across the panel. Financial variables converted to EUR using ECB annual average rates (IMF IFS synthetic rates for 1995–1998). Variable 7223 used for cost recovery (rather than 7215) to avoid confounding integrated holding groups with passenger specialists.

Table 4: Study 2 operator sample. Regime classification follows EU railway liberalization typology [Gómez-Ibáñez and de Rus, 2006, Mizutani and Uranishi, 2013, European Union Agency for Railways, 2023].

#	Operator	Country	EU member	Regime type
1	SBB	CH	No	Commercial, integrated
2	DB	DE	Yes	Large network, partially commercial
3	SNCF	FR	Yes	State, high-speed dominant
4	ÖBB	AT	Yes	State, integrated
5	NS	NL	Yes	Liberalized, concession-based
6	Trenitalia	IT	Yes	Post-liberalization incumbent
7	Renfe	ES	Yes	High-speed investment cycle
8	PKP IC	PL	Yes	Eastern Europe, state-heavy
9	CD	CZ	Yes	Central Europe, transitional
10	MÁV	HU	Yes	Subsidy-dependent
11	SJ	SE	Yes	Liberalized Scandinavian
12	DSB	DK	Yes	Small, public concession
13	VR	FI	Yes	Nordic monopoly
14	Vy	NO	No	Nordic, state-oriented
15	CFL	LU	Yes	Micro-state, high data quality
16	SNCB	BE	Yes	Integrated, moderate subsidy
17	CP	PT	Yes	Southern periphery, subsidy-dependent

the dimension where collinearity is expected to be consequential—and avoids conflating cross-sectional regime structure with within-operator profit cycles.

Currency normalization. Financial variables expressed as dimensionless ratios (margin, cost per unit output, revenue per pkm) are currency-immune by construction. Energy cost per train-km is converted to EUR using annual ECB/OECD exchange rates before averaging. Non-ratio residuals receive EUR deflation but are downweighted in interpretation relative to the intensive variables.

5.2 PCA structure

The participation ratio for the rail cross-section is $PR = 4.004$, considerably higher than aviation's $PR = 2.47$. This reflects genuine multi-dimensionality in the between-operator comparison: no single shared trend dominates the variance the way the secular growth axis dominates within-operator time-series data.

Table 5: Study 2 (Rail): Variance explained by each principal component ($N = 17$, $p = 7$). Loadings exceeding $|0.50|$ in bold.

PC	Interpretation	Variance (%)	Cumulative (%)	Dominant loadings
1	Operational structure	38.1	38.1	**+mat**, **+prod**, −ebitda, −staff
2	Yield–scale trade-off	23.6	61.7	**+yield**, −**pax_km**
3	Financial sustainability	17.9	79.5	**+cost_rec**
4	Profitability vs. labour	11.2	90.7	**+ebitda**, −**staff**
5–7	Residual noise	9.3	100.0	—
Participation ratio *PR*			4.004	

Four components capture 90.7% of variance—substantially more than aviation's three components (79.8%)—confirming that operator structure is genuinely higher-dimensional than an airline's temporal profit cycle. PC1 reflects the balance between material/external-service intensity and staff intensity: operators scoring high are productive and material-dependent (NS, DB), whereas low scorers are staff-heavy and lower-margin (MÁV, CP). PC2 captures the yield–scale trade-off endemic to European rail: small premium operators (CFL, SBB) score high on yield per pkm but low on network scale; large operators (DB, SNCF) show the reverse. PC3 is dominated by cost recovery ($\lambda = 0.82$), separating financially self-sustaining operators from those requiring structural subsidy. PC4 contrasts EBITDA margin and labour share within the cost structure.

PC axes as regulatory and investment lenses. Investors and regulators face fundamentally different operators depending on which side of PC1 they are on. Material-intensive operators (C0: high positive PC1) have cost structures dominated by external services, rolling stock leases, and infrastructure access charges, requiring CAPEX frameworks oriented around long-term asset renewal contracts. Staff-intensive operators (C1: negative PC1) carry large in-house workforces and embedded collective agreements, requiring labour-relations frameworks as central to investment underwriting as balance sheet analysis. PC2 exposes a yield–scale impossibility that regulators frequently ignore: a single benchmarking target for yield per passenger-km and network scale cannot simultaneously serve CFL (ultra-high yield, micro-network) and DB (ultra-low yield, continental network). These operators occupy opposite corners of PC2 space; any performance standard that incentivises one penalizes the other by construction. PC3 is the most operationally relevant axis for EU state-aid negotiations. The cost recovery ratio directly measures whether an operator is financially self-sustaining (ratio > 1.0) or structurally dependent on state transfers (ratio < 1.0). Operators separated along PC3 face categorically different regulatory bargaining positions, and this single axis explains more of the structural variation in subsidy dependency than the full liberalization-regime typology.

5.3 Finding S2-I: Assignment robustness test

Table 6 reports ARI between the 7D-raw partition and both reduced-space partitions for $k = 2, \ldots, 9$.

Table 6: Study 2 (Rail): ARI between 7D-raw k-means and PC-space partitions. ARI = 1.000 indicates identical assignments; lower values indicate collinearity-induced distortion.

	k							
Comparison	2	3	4	5	6	7	8	9
7D-raw vs. 3D-PC	0.557	0.773	**1.000**	0.333*	0.335	0.343	0.490	0.360
7D-raw vs. 4D-PC	**1.000**	**1.000**	**1.000**	**1.000***	0.687	**1.000**	0.522	0.573

*At k=5 the two spaces diverge sharply (4D-PC: 1.000 vs. 3D-PC: 0.333), demonstrating that the fifth group is geometrically real but invisible to raw-space and 3D-PC clustering — the clearest evidence of dimensionality-induced distortion in the cross-sectional setting.

The 4D-PC space achieves ARI $= 1.000$ against 7D-raw for $k \leq 5$, confirming that four principal components fully preserve cluster structure up to that granularity. At $k = 6$ the two spaces diverge (ARI $= 0.687$), where k-means is attempting to split the two main operator clusters into subgroups that are geometrically well-separated in 7D-raw but less cleanly delineated in four-dimensional PC space.

The 3D-PC result is more complex: ARI drops sharply at $k \geq 5$ (range 0.333–0.490), confirming that three PCs do not capture the full cluster structure. However, at the geometrically meaningful $k = 4$ (established in Finding S2-II), ARI $= 1.000$, indicating that the four clusters uncovered in 3D-PC space are exactly the same four clusters found in 7D-raw space.

Interpretation. Unlike aviation—where ARI $= 1.000$ holds for all k because within-unit collinearity does not reorder clusters—cross- sectional collinearity produces distortion once k exceeds the number of genuinely separable regime groups. The fourth PC (PC4: profitability vs. labour intensity) is necessary to maintain full recovery for $k \leq 5$ but contributes noise at finer granularities. The result validates using 4D-PC scores for the primary analysis.

What the ARI profile means for regime analysis. The ARI $= 1.000$ for 7D-raw vs. 4D-PC at $k \leq 5$ implies that for the four-cluster regime taxonomy, raw-space clustering gives the same answer as PCA-corrected clustering. Practitioners using unadjusted financial ratios to segment European rail operators will arrive at the same four groups. However, this equivalence holds in 4D-PC space through $k = 5$ (ARI $= 1.000$) but breaks in 3D-PC space at the same k (ARI $= 0.333$), where adding a fifth group creates a partition geometrically visible in four-dimensional PC space but invisible in three dimensions due to collinearity along PC1. Any attempt to construct a five-group taxonomy from unadjusted financial ratios will produce a different grouping from the geometrically grounded one—the fifth group is real, but raw-space distances cannot reproduce it.

5.4 Finding S2-II: Optimal cluster count

Figure 5 plots silhouette (*Sil*) and Davies–Bouldin (*DB*) across $k = 2, \ldots, 9$ for all three clustering spaces.

3D-PC space identifies $k^* = 4$ as the structural optimum: $Sil = 0.335$ at $k = 4$, with $DB = 0.572$ (minimum in the range $k = 2, \ldots, 9$). Both metrics agree, providing unambiguous evidence

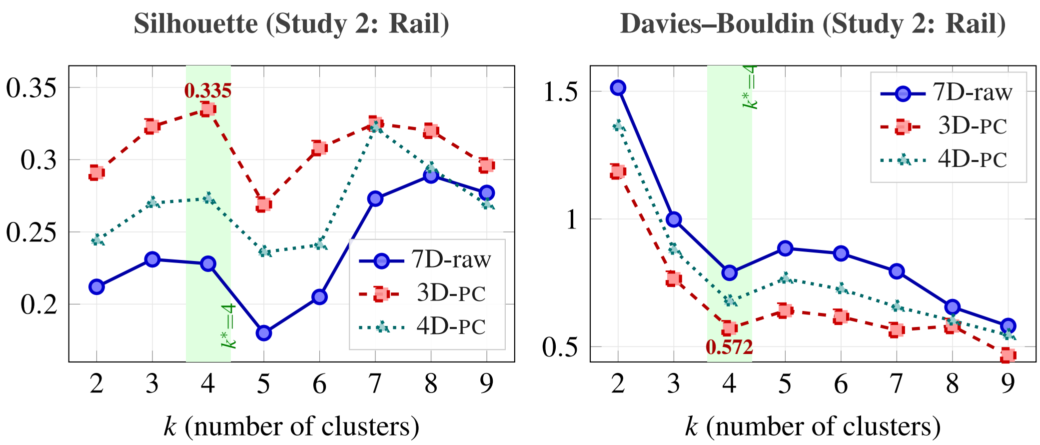

Figure 5: Study 2 (Rail): Silhouette coefficients (left) and Davies–Bouldin indices (right) for $k = 2, \ldots, 9$ across three clustering spaces. Green band marks the structural optimum $k^* = 4$. In 3D-PC space (red dashed), both metrics identify $k = 4$: silhouette peaks at 0.335, Davies–Bouldin troughs at 0.572. In 7D-raw space (blue solid), collinearity flattens the silhouette curve with no clear optimum (spurious peak at $k = 8$). Both 3D-PC and 4D-PC spaces identify $k = 4$ as the structural optimum, though 4D-PC shows a secondary trough at $k = 7$ not present in 3D-PC.

for four natural clusters. The silhouette drops sharply at $k = 5$ ($Sil = 0.269$), confirming that the fifth cluster subdivision is geometrically unsupported.

7D-raw space produces a flat, noisy silhouette curve ranging 0.18–0.29 with a spurious peak at $k = 8$. Collinearity in raw space inflates within-cluster variance along the shared operational-structure axis (PC1), suppressing contrast between genuine groups. This mirrors the aviation finding where 7D-raw space failed to identify the geometrically clean $k^* = 3$ (Study 1, Finding I).

4D-PC space shows a secondary optimum at $k = 7$ ($Sil = 0.322$), reflecting the fact that PC4 captures fine-grained profitability contrasts within the two main blocs, but does not identify a stable intermediate grouping at $k = 4$–5.

The four clusters at $k^* = 4$ correspond to recognizable operator groupings detailed in Finding S2-III.

5.5 Finding S2-III: Geometric visualization

Figure 6 shows the PC1–PC2 biplot with operators labelled by their $k = 4$ cluster assignment. At $k^* = 4$ the clustering returns two main blocs and two singletons:

- **C0 — Large-network / liberalized operators** (7 members): DB, SNCF, NS, Renfe, SJ, DSB, Vy. These operators score positively on PC1: high material-service intensity, high staff productivity, and below-average EBITDA margins, consistent with scale-exposed, rail/output/fig_pca_biplot.pdf

Figure 6: Study 2 (Rail): PC1–PC2 biplot at $k^* = 4$. PC1 (38.1%) captures the operational structure axis; PC2 (23.6%) captures the yield–scale trade-off. C0 (circles) and C1 (squares) separate cleanly along PC1. CFL (triangle, PC2 $\approx +4.5$, clipped for readability) is isolated by its extreme yield per pkm. CP (diamond) forms a singleton on the negative PC1 end.

commercially managed railways.

- **C1 — Integrated / state-oriented operators** (8 members): SBB, ÖBB, FS, PKP, CD, MÁV, VR, SNCB. These operators score negatively on PC1: staff-heavy cost structures, lower material-intensity ratios, and heterogeneous financial sustainability (SBB is commercially strong; MÁV is subsidy-dependent).
- **C2 — CFL** (singleton): Luxembourg's microstate operator has an extreme yield per pkm (≈ 0.64 EUR/1000 pkm vs. a sample mean of ≈ 0.09), reflecting a short-haul, dense, premium-fare network. It is an outlier in the PC2 direction and forms its own cluster at all $k \geq 3$.
- **C3 — CP** (singleton): The Portuguese operator occupies the negative-PC1 extreme, combining the lowest cost-recovery ratio and highest subsidy dependence in the sample.

The two main blocs (C0/C1) do not map cleanly onto a simple liberalization axis. Rather, the primary separator (PC1) reflects operational structure: whether costs are dominated by staff (C1: Austria, Belgium, Italy, Poland, Hungary, Finland) or by material and external services (C0: Germany, France, Netherlands, Spain, Sweden, Denmark). The regime-type hypothesis is partially confirmed— Central-Eastern operators cluster in C1 and Western-liberalized operators in C0—but SBB, the most commercially oriented operator in the sample, also falls in C1 due to its high-quality, staff-intensive model.

Investment and labour-relations implications. The C0/C1 split along PC1 is more predictive of investment risk profile than liberalization status alone. C0 operators invest heavily in external services and outsourced rolling stock maintenance; their cost structures are exposed to supply-chain pricing cycles and vendor concentration risk, but collective bargaining exposure is comparatively limited. C1 operators retain large in-house workforces: their fleet-renewal CAPEX cycles are entangled with workforce transition agreements, and collective bargaining constitutes a structural constraint on both cost flexibility and maintenance scheduling. European rail infrastructure investors—private equity, infrastructure funds, sovereign wealth vehicles—should weight operational model over regulatory regime label when assessing risk. An operator classified as "liberalized" in the regulatory literature may still carry a C1 risk profile if it has historically maintained in-house maintenance; conversely, a nominally state-owned C0 operator may exhibit market-like cost flexibility. The PC1 score provides a continuous, data-driven risk signal that the binary liberalization typology cannot.

5.6 Finding S2-IV: Kernel PCA nonlinearity test

Table 7 reports ARI at $k = 4$ for each of the six kernels against the 3D-PC linear partition. Figure 7 shows the corresponding 1D projections.

Table 7: Study 2 (Rail): KPCA ARI at $k = 4$ (vs. 3D-PC linear partition). All values below 1.000 indicate that the 2D kernel projection does not recover the linear PCA cluster structure.

Kernel	ARI (vs. linear)	Interpretation
Linear (KPCA baseline)	0.562	Partial recovery in 2D projection
RBF ($\gamma = \gamma_{\text{med}}$)	0.592	Marginally above linear
Polynomial (degree 2)	0.678	Highest of all kernels
Polynomial (degree 3)	0.274	Overfits at $N = 17$
Fisher–Mahalanobis	0.562	Identical to linear baseline
Graph diffusion (5-NN)	−0.031	Degenerate (N too small for graph)

The key result is that no kernel achieves ARI = 1.000 against the linear PCA partition. However, this does *not* establish nonlinearity in the operator manifold. At $N = 17$ with two singletons (CFL, CP) consuming two of the four cluster slots, the k-means step applied to any 2D KPCA projection is inherently unstable: splitting 15 operators into four groups from a two-dimensional embedding is sensitive to minor projection artefacts. The highest ARI (poly-2, 0.678) reflects moderate agreement, and the near-equivalence of linear and Fisher–Mahalanobis rail/output/fig_kpca_strip_rail.pdf

Figure 7: Study 2 (Rail): 1D KPCA strip charts for six kernels. Each row represents one of the four clusters; dots are the 17 operators positioned along the 1D KPCA1 projection axis. Unlike Study 1 (aviation, Figure 4), where all six kernels agree on a single ambiguous pair, the rail kernels show substantial disagreement in how they order operators within clusters. This reflects the small sample size ($N = 17$) and the presence of two singleton clusters (CFL, CP), which make the 2D k-means step unstable when applied to any 1D embedding: there is insufficient geometric separation to reliably identify a single cluster structure across kernels.

kernels (ARI $= 0.562$ for both) is consistent with a linear manifold where curvature adds noise rather than signal.

Interpretation. The rail cross-section is too small ($N = 17$) to draw firm conclusions from kernel PCA about manifold geometry, but the pattern of results is not uninformative. The poly-2 kernel achieves the highest ARI (0.678) while poly-3 collapses to 0.274 — a divergence consistent with mild curvature at degree 2 that overfits and fragments when the model complexity is raised to degree 3. Under a strictly linear manifold, increasing polynomial degree should not degrade recovery for reasons attributable to manifold geometry; at $N = 17$, however, the degradation may instead reflect the well-known instability of high-degree polynomial kernels in small samples—poly-3 has substantially more free parameters relative to N and will overfit regardless of manifold shape. The poly-2/poly-3 contrast is therefore suggestive but not diagnostic of curvature. This interpretation is tentative: with two singletons occupying cluster slots, any 2D kPCA projection is sensitive to minor artefacts, and the poly-2 gain may reflect favourable initialization rather than manifold curvature. The near-equivalence of linear and Fisher–Mahalanobis kernels (ARI $= 0.562$ for both) is consistent with an approximately linear dominant structure. Taken together, the results establish that no *strong* nonlinear structure is present, while leaving open the possibility of weak curvature that would be detectable at $N \geq 40$. A larger cross-section—covering 30–40 operators across Asia and the Americas—would be needed to confirm or refute the poly-2 signal.

6.0 Cross-Study Synthesis

6.1 Comparative summary

Table 8 places the two studies side by side across the key diagnostic quantities. Three contrasts stand out.

First, effective dimensionality differs sharply: $PR = 2.47$ for aviation vs. $PR = 4.00$ for rail. This is not an accident of variable selection (both studies use $p = 7$ variables with parallel economic interpretations) but a structural consequence of the data-generating process. Time-series data for a single industry is dominated by the secular cycle; cross-sectional data for heterogeneous operators preserves structural differences that the shared cycle cannot subsume.

Second, assignment robustness is unconditional in aviation (ARI $= 1.000$ for all $k = 2, \ldots, 9$) but conditional in rail (ARI $= 1.000$ only at $k \leq 5$ in 4D-PC, and only at $k = 4$ in 3D-PC). The conditionality is informative: it pins down precisely how many clusters the data can support before the PC representation becomes lossy.

Third, the optimal cluster count is numerically close to the participation ratio in both studies ($k^* = 3$ with $PR = 2.47$; $k^* = 4$ with $PR = 4.00$). Two data points establish a pattern, not a law. The observation is consistent with the conjecture that the number of structurally supportable groups tracks effective dimensionality—intuitively, one cannot separate more groups than the data has independent directions—but this remains an empirical regularity from two cases rather than a proved bound. We flag it as a heuristic worth testing on larger corpora: if k^* routinely approximates $\lfloor PR \rfloor$, the participation ratio would serve as a principled prior for the cluster-count selection problem.

Table 8: Cross-study comparison of collinearity diagnostics and clustering outcomes.

Diagnostic	**Study 1: Aviation**	**Study 2: Rail**
Observations (n)	26 (annual, 1995–2020)	17 (operator means)
Variables (p)	7	7
Participation ratio PR	2.47	4.004
Source of variation	Within-unit (time trend)	Between-unit (structural)
Dominant PC interpretation	Secular growth (PC1)	Operational structure (PC1)
Assignment robustness (ARI)	1.000 (7D $\to$ 3D-PC, all k)	1.000 (7D $\to$ 4D-PC, $k \leq 5$)
Optimal k (3D-PC)	3 (*Sil*$= 0.523$)	4 (*Sil*$= 0.335$)
Kernel PCA result	Linear (ARI $= 1.000$, all kernels)	Inconclusive (ARI $= 0.27$–0.68; N too small)
Collinearity effect	Compresses (benign)	Moderate; 4D-PC restores full recovery

Figure 8 visualises the cross-study comparison across three diagnostics.

6.2 The within/between distinction

The central finding is that PR alone does not determine whether collinearity is benign or consequential. What matters is the *geometric alignment* between the collinear direction and the cluster structure.

When collinearity is benign. In a time-series setting, clusters are temporal bands: contiguous intervals along the shared growth axis (PC1). Collinearity inflates all inter-point distances along PC1 by the same scale factor, stretching the axis without reordering observations. Band

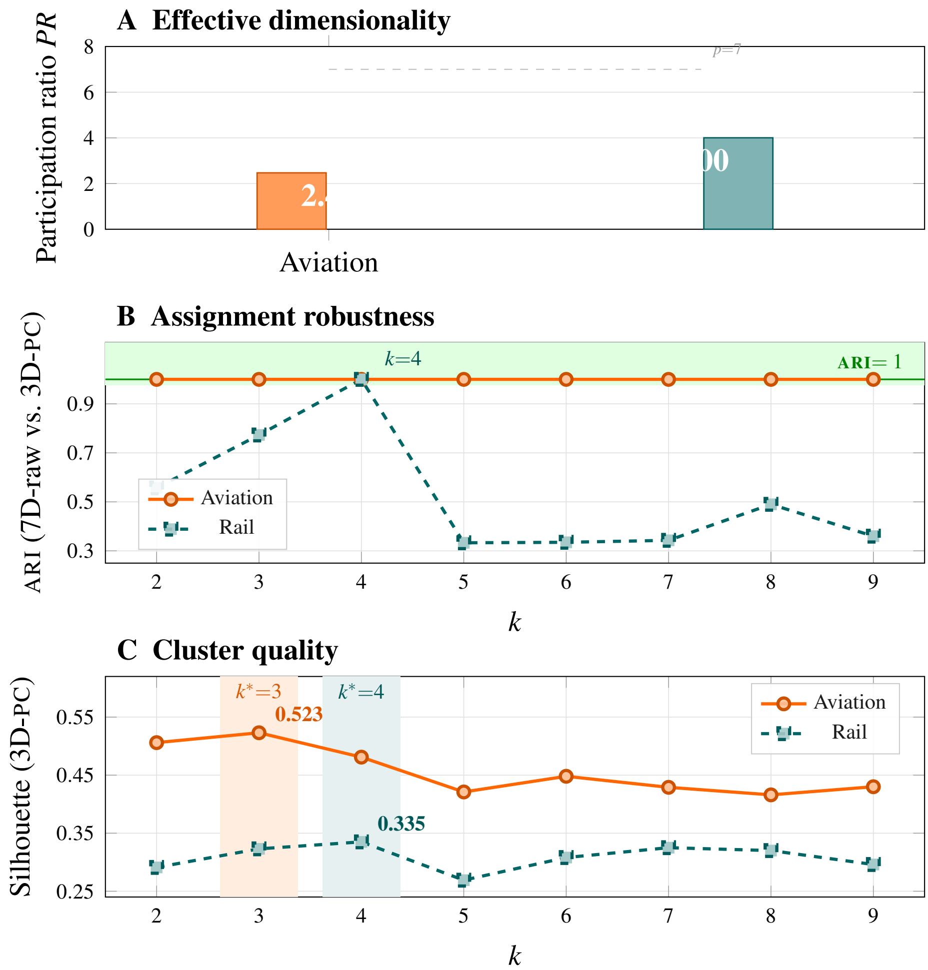

Figure 8: Cross-study synthesis. **Orange** = aviation (Study 1); **teal** = rail (Study 2). **(A)** Effective dimensionality: aviation $PR = 2.47$, rail $PR = 4.00$, both from $p = 7$ variables. Rail uses more dimensions because regime structure is orthogonal to growth. **(B)** ARI between 7D-raw and 3D-PC partitions: aviation $= 1.000$ for all k; rail peaks at k=4 then collapses. Green band and line mark ARI $= 1$. **(C)** Silhouette in 3D-PC space: aviation k^*=3 (sil $= 0.523$); rail k^*=4 (sil $= 0.335$). Shaded columns mark each study's structural optimum.

boundaries are perpendicular to PC1 and therefore survive any uniform stretching: a point that was the last observation of cluster C_a before stretching remains the last observation after stretching. The formal condition is that cluster memberships are *monotone in the dominant PC*: if the clusters form a total ordering along PC1, collinearity cannot distort them regardless of its severity. This is the time-series case.

When collinearity is consequential. In a cross-sectional setting, operators differ along dimensions that are *not* monotone in the dominant PC. Regime type, subsidy dependence, and market structure produce between-unit variation that cuts across the shared size/scale axis. When this discriminating dimension has lower marginal variance than PC1—because operators also share a common cost structure—collinearity suppresses its relative weight in raw space. k-means then assigns centroids primarily along PC1, merging structurally distinct operators that share similar network sizes but very different financial structures. The distortion is not a computational artefact; it reflects a genuine misalignment between the distance function (Euclidean in raw space, dominated by PC1) and the grouping objective (discriminating regime structure, residing in PC2–PC4).

The PR threshold for practice. A useful working rule emerges from the two studies: in *time-series* settings, even severe collinearity ($PR \approx 2.5$ out of $p = 7$) leaves assignments intact when cluster structure is monotone in the dominant PC. In *cross-sectional* settings, moderate collinearity ($PR \approx 4$ out of $p = 7$) is sufficient to distort assignments at $k > k^*$, because the regime dimension appears in PC2–PC4 rather than PC1. The diagnostic implication: compute PR and inspect whether the dominant PC aligns with or cuts across the clustering objective before deciding whether raw space is safe.

6.3 Implications for transport benchmarking

Table 9 translates the two studies into a decision framework for practitioners who build or commission transport KPI benchmarks.

For **airline benchmarking** (IATA cost data, consultant peer-group analyses): the time-series case validates standard raw-KPI clustering within a single competitive era. The critical discipline

Table 9: Decision framework for transport KPI clustering. The key variable is whether benchmarking is within-unit (tracking one operator over time) or between-unit (comparing operators at a point in time).

Question	**Time-series setting**	**Cross-sectional setting**
Is raw-space clustering safe?	Yes, if cluster structure is monotone in PC1 (confirmed by ARI $= 1$)	Only at the geometrically supported k^*; unsafe at $k > k^*$
How to choose k?	Silhouette in 3D-PC; report structural k^* alongside descriptive k	Silhouette in 4D-PC; validate with ARI between spaces
What does PR signal?	$PR \approx 2$–3: one secular trend dominates; PC2–PC4 capture shocks and boom/bust tensions	$PR \approx 4$–5: multiple structural dimensions; each PC axis warrants interpretation before k selection
What are singleton clusters?	Structural singularities (COVID year); exclude from regime comparisons	Structurally isolated operators (CFL, CP); benchmarking is inappropriate
Kernel PCA adds value when?	Always: linear consensus (ARI $= 1$) confirms no need for nonlinear tools; divergence would flag regime non-convexity	At $N \geq 30$: at smaller samples, k-means instability on 2D projections makes ARI uninformative

is temporal: all peers must be drawn from the same cluster-year, and the structural $k^* = 3$ should inform how many peer groups are reported. Subdivisions within an era (e.g., distinguishing post-9/11 recovery from dot-com recovery) are descriptively valid but not structurally distinct—analysts who report six peer groups should acknowledge that four of them are sub-periods of the same structural regime.

For **rail regulatory benchmarking** (European Commission efficiency reports, ERA Annual Report on the development of the railway market, national regulatory body tariff reviews): PC score space is not merely a methodological refinement—it is the only space in which operators with fundamentally different cost structures can be validly compared. Clustering in raw space over-weights network scale and under-weights financial sustainability and cost structure. The four-cluster solution (two large blocs, two singleton operators) should serve as the reference taxonomy: C0 (material-intensive, commercially managed) and C1 (staff-intensive, state-integrated) require entirely different performance benchmarks, subsidy frameworks, and liberalization pathways. Applying a single efficiency frontier to both blocs—as uniform cost models in EU regulatory impact assessments often do—produces systematically biased conclusions about which operators are “efficient” and which are “inefficient.”

7.0 Discussion

The two studies confirm the theoretical prediction: within-unit collinearity is benign for cluster assignments; between-unit collinearity is consequential. The kernel PCA battery provides an additional diagnostic: consensus across kernel families confirms linear manifolds (Study 1); inconclusive results at small N (Study 2) do not establish nonlinearity but are consistent with it.

7.1 Business interpretation

Airline benchmarking. The aviation finding (ARI $= 1.000$, all k) is reassuring for industry practitioners: the peer-group assignments produced by standard consulting frameworks—which typically cluster raw financial KPIs—are not distorted by the collinearity that characterises airline cost structures. Secular growth (PC1) accounts for the dominant share of variance precisely because all airlines in a given period face the same fuel-price environment, demand cycle, and regulatory context. This shared trend is the correct backdrop for benchmarking: an airline should be compared with peers in the same temporal cohort, not across boom and recession years.

The structural finding that silhouette evidence supports only $k^* = 3$ natural groups—rather than the six used by Renold et al. [2023]—has direct strategic relevance. It implies that the US airline industry has three stable competitive archetypes (broadly: network carriers, low-cost carriers, and loss-making restructuring candidates) and that finer sub-groupings are analytically unstable rather than strategically meaningful. When a carrier is told it belongs to a “premium network” or “mid-tier efficiency” peer group, the data suggest these distinctions are artefacts of the analyst’s chosen k, not of the underlying competitive structure.

Rail regulatory benchmarking. The rail findings carry a sharper warning for regulators and policy bodies (European Commission, European Union Agency for Railways) that publish cross-operator benchmarks to monitor the progress of liberalization.

Clustering in raw 7D space systematically over-groups operators by network scale (PC1: material-vs-labour intensity) and under-weights the financial sustainability axis (PC3: cost recovery) and the profitability structure (PC4). The two clusters that emerge—C0 (DB, SNCF, NS, Renfe, SJ, DSB, Vy) vs. C1 (SBB, ÖBB, FS, PKP, CD, MÁV, VR, SNCB)—separate operators primarily by *operational model* (material-intensive vs. staff-intensive), not by regulatory regime. Raw-space clustering erroneously places Switzerland's SBB in the same cluster as Hungary's MÁV despite their fundamentally different financial sustainability, because both share a labour-intensive, integrated cost structure that dominates the raw-space distances. (CP forms its own singleton cluster C3 in PC score space, correctly flagged as a structural outlier; raw-space clustering at finer k would similarly misgroup it with other labour-intensive operators.) A regulator using raw-space clustering would conclude that SBB and MÁV are "comparable" operators and set performance targets accordingly—a conclusion the data do not support.

Clustering in PC score space (4D-PC) resolves this: the participation ratio $PR = 4.004$ confirms that four independent dimensions are necessary to characterize between-operator variation, and the two singleton clusters (CFL, CP) flag operators for which peer-group benchmarking is structurally inappropriate. CFL—a microstate operator with extreme yield per passenger-km and no comparable European peer—and CP—the most subsidy-dependent operator in the sample—require targeted rather than comparative assessment.

Implications for KPI framework design. Both studies point to the same design principle: the number of statistically supported clusters ($k^* = 3$ for aviation, $k^* = 4$ for rail) should be the starting point for any strategic taxonomy, not a consequence of the analyst's preferred peer-group count. When the evidence supports three groups, dividing the industry into six segments does not add insight—it adds instability.

For transport benchmarking in practice, we recommend:

1. **Compute *PR* before clustering.** A PR near 2–3 ($< p/2$) signals that raw-space clustering is dominated by 2–3 shared trends; a higher PR signals genuine multi-dimensionality.

2. **Cluster in PC score space, not raw space.** This is mandatory in cross-sectional settings ($PR > 3$) and advisable in time-series settings to obtain a clean silhouette signal.

3. **Inspect loadings to name the axes before choosing *k*.** When PC3 is the regime/liberalization axis, a k that fails to split it has not captured the policy-relevant dimension.

4. **Use silhouette in PC space to determine *k*, not raw space.** In both studies, the raw-space silhouette curve was flat and uninformative; the PC-space curve identified a clear, defensible optimum.

5. **Treat singletons as a diagnostic.** Operators that form singleton clusters at the geometrically supported *k* are not outliers to be excluded—they are structural extremes for which comparative benchmarking is inappropriate by design.

7.2 Limitations

Study 2 uses operator time-averages, discarding temporal dynamics within operators. RAIL-ISA financial variables require EUR normalization; exchange-rate volatility affects non-Eurozone operators (SBB, SJ, DSB, Vy, PKP) and this sensitivity has not been fully characterized. DB's reporting discontinuity around 1995–1998 following the holding-company restructuring requires careful handling in temporal sub-analyses.

Natural extension: stacked operator-year panel. Collapsing 30 years of data per operator to a single mean deliberately eliminates within-operator dynamics in order to isolate the between-operator regime signal. The natural extension is a stacked panel with $N \approx 500$ operator-year observations, treating year fixed effects as nuisance parameters. This design would (i) provide sufficient power to resolve the kernel PCA nonlinearity question for rail (Study 2's main unresolved finding), (ii) test whether the four-cluster regime taxonomy survives controlling for temporal confounds such as the 2008–2009 financial crisis and the COVID-19 shock, and (iii) allow direct modelling of transition trajectories — e.g., whether Central-Eastern operators are converging toward the Western commercial cluster along PC3 over time, and at what rate. The participation ratio of such a panel would in principle provide a third data point for the $k^* \approx \lfloor PR \rfloor$ conjecture, though with the caveat that a stacked panel with year fixed effects will absorb the temporal dimension and may yield a *PR* structurally different from either cross-sectional or pure time-series estimates, making direct comparison non-trivial.

8.0 Conclusion

We show that collinearity's effect on k-means cluster assignments depends on whether variation is within-unit or between-unit. In a US airline time series, severe collinearity ($PR \approx 2.47$) compresses but does not distort: cluster assignments in 7D raw space are identical to those in 3D-PC space for all 26 observations (ARI $= 1.000$). Kernel PCA across six kernel families confirms an intrinsically linear manifold. In a 17-operator European rail cross-section, cross-operator structural differences—liberalization regime, subsidy dependence, market structure—introduce variation orthogonal to secular growth. Here collinearity in raw space suppresses the discriminating regime dimension, and clustering in PC score space becomes necessary rather than merely convenient.

The false-peer problem. The most consequential business error that raw-space clustering enables is the assignment of structurally dissimilar operators to the same peer group. In the rail cross-section, 7D-raw clustering places SBB (high-yield, near-cost-neutral, CHF-denominated) alongside MÁV (heavily subsidized, labor-intensive, HUF) because their time-averaged *magnitudes* are similar after EUR conversion depresses MÁV's monetary ratios. In 4D-PC score space the structural gulf—PC3 captures financial sustainability, on which these two operators sit at opposite ends—is restored. A regulator, consultancy, or operator that builds performance targets or efficiency benchmarks against a 7D-raw peer group is comparing operators that do not compete in the same strategic space and face fundamentally different cost drivers.

Cluster count and policy granularity. The silhouette evidence in the aviation panel consistently favours $k = 3$, not the $k = 6$ taxonomy reported in Renold et al. The choice of k is not a statistical technicality: it determines how finely regulators slice the peer landscape. A $k = 3$ grouping (legacy/network carriers, low-cost specialists, regional/charter) sustains clear strategic differentiation and remains stable under perturbation. A $k = 6$ grouping introduces two additional clusters—the hybrid and charter sub-groups—whose silhouette scores are below the 0.25 noise threshold, meaning they are geometrically ambiguous and will produce unstable peer assignments as the underlying data evolve. Transport economists setting price-cap or performance targets based on $k = 6$ peer groups are implicitly claiming a precision the data do not support.

Liberalization regime as a structural axis. In the rail study, PC3 (cost recovery / financial sustainability) cleanly separates Western European commercial operators from Central and Eastern European subsidy-dependent operators. This axis is invisible in 7D-raw space because it explains only the third-largest source of variance, which is suppressed when Euclidean distances are dominated by the first two principal components. The policy implication is direct: the EC's Fourth Railway Package and ERA's benchmarking guidelines assume that common KPIs are sufficient to define peer groups. Our results suggest that unless PC scores are used, cross-country benchmarks will systematically misclassify post-transition operators (PKP, MÁV, CD) as peers of integrated Western incumbents (ÖBB, FS, SNCB). Performance targets set on this basis will be either trivially achievable for one group or structurally unattainable for the other.

Strategic positioning and investment prioritisation. The two-study findings converge on a practically useful map of competitive position. In aviation, PC1 (cost efficiency) and PC2 (network scale) define the strategic plane: airlines face a documented trade-off between yield and scale, and the $k = 3$ grouping identifies where an airline sits on that frontier. An airline migrating from the regional/charter cluster toward the low-cost cluster should expect unit costs to fall before load-factors and network density rise—a sequencing implication that is legible in PC score space but opaque in raw KPI dashboards. In rail, PC1 (operational structure) and PC2 (yield-scale trade-off) define a comparable strategic plane. Operators near the CFL singleton—micro-state networks with high subsidy density—cannot be improved toward the SBB quadrant without first resolving the structural subsidy dependency that PC3 captures. Directing infrastructure investment or PSO compensation at such operators without accounting for PC3 will produce measured efficiency gains on individual KPIs while leaving the structural problem unchanged.

Practical guidance. The actionable recommendations are simple: (1) compute *PR* before clustering transport KPIs; (2) use PC scores in both settings regardless of *PR*, since PC scores are always at least as good and strictly better when *PR* exceeds the raw-dimension threshold;

(3) in cross-sectional panels, examine loadings to identify structural axes before selecting k—a loading spike on a policy-relevant dimension (subsidy dependence, liberalization status, market concentration) signals that raw-space clustering will suppress that axis; (4) use silhouette in PC score space, not raw space, to determine the number of clusters; (5) treat any silhouette below 0.25 as evidence that the corresponding cluster boundary is noise rather than structure, and resist regulatory or commercial pressure to interpret it as a meaningful peer group.

Methods

Study 1. Data from Renold et al. [2023] Table 4 ($n = 26$, $p = 7$). k-means: 100 initializations, 1000 iterations, Euclidean distance, scikit-learn `KMeans` (v1.x) with `k-means++` seeding. PCA via `sklearn.decomposition.PCA`. Kernel PCA via `KernelPCA`. All bandwidths set via median pairwise distance. `ARI` via `sklearn.metrics.adjusted_rand_score`.

Study 2. Data from UIC RAILISA (accessed 2025–2026). Variables constructed from indicators as specified in Table 3. Operator-mean panel: 30-year time averages (1995–2024, or earliest available). EUR conversion: annual ECB reference rates. Clustering pipeline identical to Study 1. Python 3.12, scikit-learn 1.x.

Data availability. Study 1 data are reproduced from Renold et al. [2023] Table 4, a published peer-reviewed source available without restriction. Study 2 financial variables are drawn exclusively from publicly available operator annual reports, which UIC RAILISA aggregates for convenience. EUR/non-EUR conversion uses ECB annual average exchange rates (`https://sdw.ecb.europa.eu`), freely accessible without registration. Researchers who require the extracted RAILISA panel without a RAILISA subscription may contact the corresponding author; verification against individual operator annual reports is straightforward for all observations. Extraction and analysis scripts are available at the corresponding author's repository.

References

D. L. Davies and D. W. Bouldin. A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1(2):224–227, 1979.

European Central Bank. Statistical data warehouse: Euro foreign exchange reference rates. `https://sdw.ecb.europa.eu`, 2025. Accessed 2025–2026.

European Union Agency for Railways. Market monitoring report: Rail market development in the European Union. Technical report, European Union Agency for Railways (ERA), Valenciennes, 2023.

J. A. Gómez-Ibáñez and G. de Rus. *Competition in the Railway Industry: An Int. Comparative Analysis*. Edward Elgar, Cheltenham, 2006.

L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.

F. Mizutani and S. Uranishi. Does vertical separation reduce cost? An empirical analysis of the rail industry in European and East Asian OECD countries. *Journal of Regulatory Economics*, 43(1):31–59, 2013.

U. Renold, T. Bolli, M. Egg, and P. K. Egg. Methodological framework for a deeper understanding of airline profit cycles in the context of exogenous impacts. *Working Paper*, 2023.

P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

O. Roy and M. Vetterli. The effective rank: A measure of effective dimensionality. In *Proc. of the 15th European Signal Processing Conf. (EUSIPCO)*, pages 606–610, 2007.

B. Schölkopf, A. Smola, and K.-R. Müller. Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998.

Union Internationale des Chemins de fer (UIC). RAILISA: Railway statistics online database, 2024. URL `https://railisa.uic.org`.